Lab 07 – Exploring Internet of Things with Cisco Packet Tracer

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INTRODUCTION

This lab report documents the exploration and configuration of Internet of Things (IoT) devices using Cisco Packet Tracer. The primary objective was to gain hands-on experience deploying, connecting, and managing a range of IoT devices within simulated smart home environments. Activities progressed from basic device registration to custom device creation and script modification, culminating in a comprehensive final assessment.

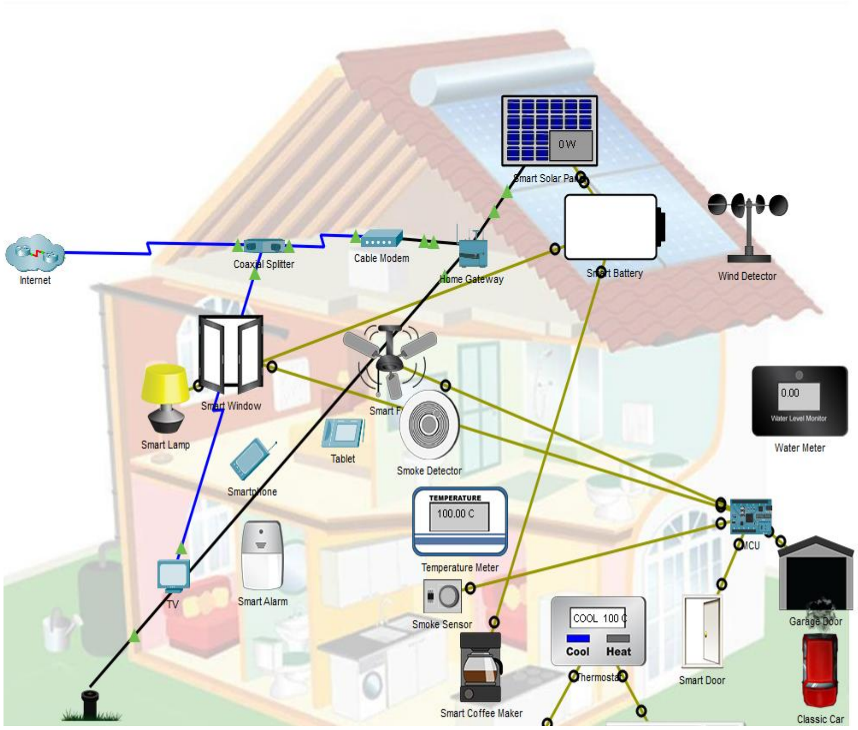
MODULE 1: CREATE YOUR OWN SMART HOME NETWORK

Module 1 provided a structured introduction to smart home networking concepts, offering an overview of the course activities and the various network-connected devices found in modern residential environments. Supplementary resources were included to establish foundational knowledge prior to beginning the hands-on activities.

1.1 IoT Devices in Packet Tracer

This section introduced the IoT device catalog available within Cisco Packet Tracer through instructional content and a supporting video overview. The video demonstrated core IoT concepts, including device types, communication protocols, and practical use cases within a networked home environment.

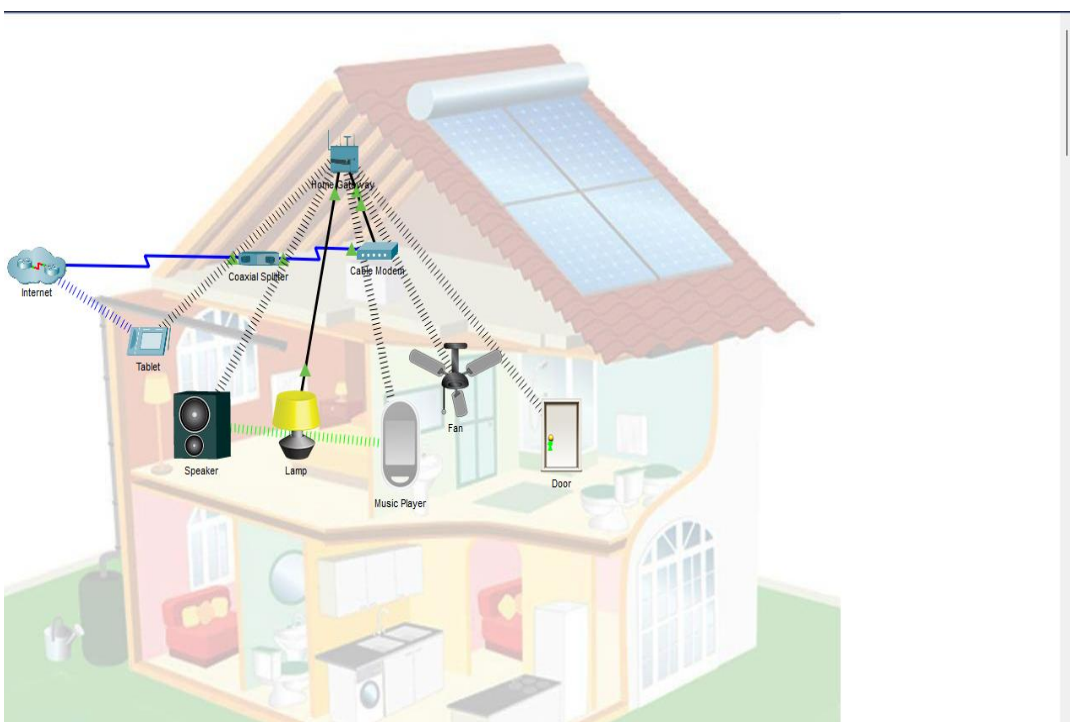
During the corresponding activity, three IoT devices were deployed and configured within Packet Tracer: a lawn sprinkler, a wind detector, and a water level monitor. Each device was successfully registered and linked to the home gateway, establishing centralized network management for the simulated environment.



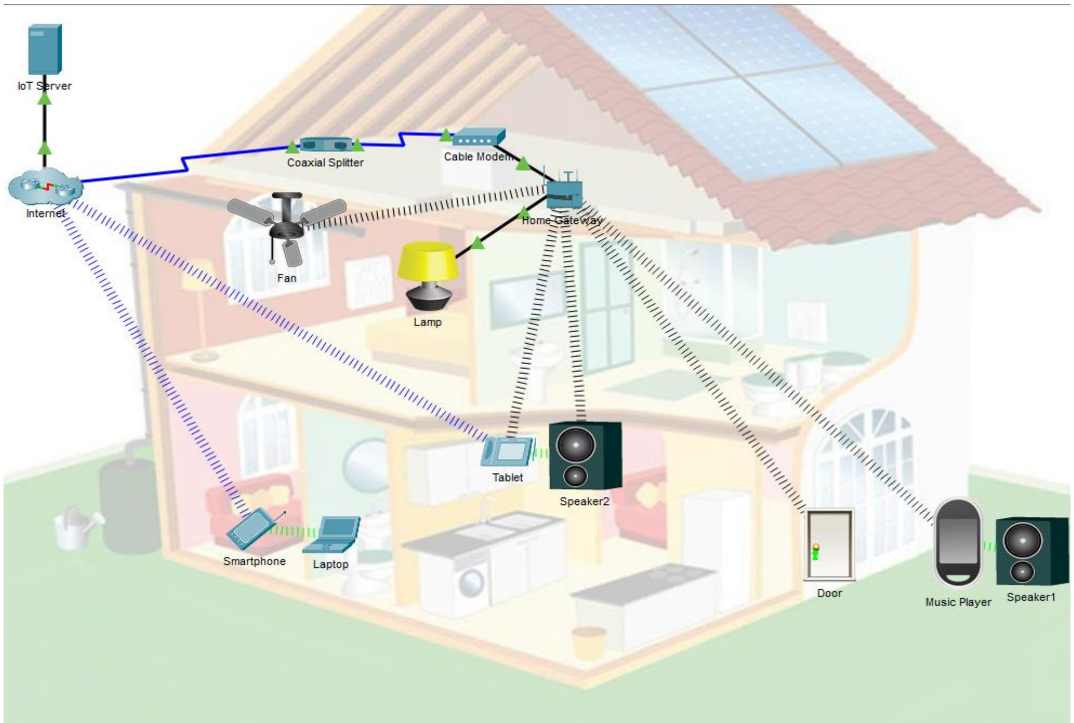
1.2 Create and Control Your Smart Home Network

This section extended the concepts introduced in Section 1.1 by demonstrating how to actively monitor and control IoT devices through the Cisco Packet Tracer interface. An instructional video outlined the monitoring workflow, emphasizing real-time device status and control panel navigation.

In the first activity, additional IoT devices were integrated into the smart home topology, including Bluetooth speakers, portable speakers, smart doors, and ceiling fans. All devices were connected wirelessly and managed through a newly configured home gateway, reinforcing the principles of centralized wireless device control.



The second activity introduced registration server-based device management. Devices were connected to both wireless and Bluetooth networks, a laptop and smartphone were paired via Bluetooth, and the laptop was tethered to the smartphone to simulate mobile connectivity scenarios.

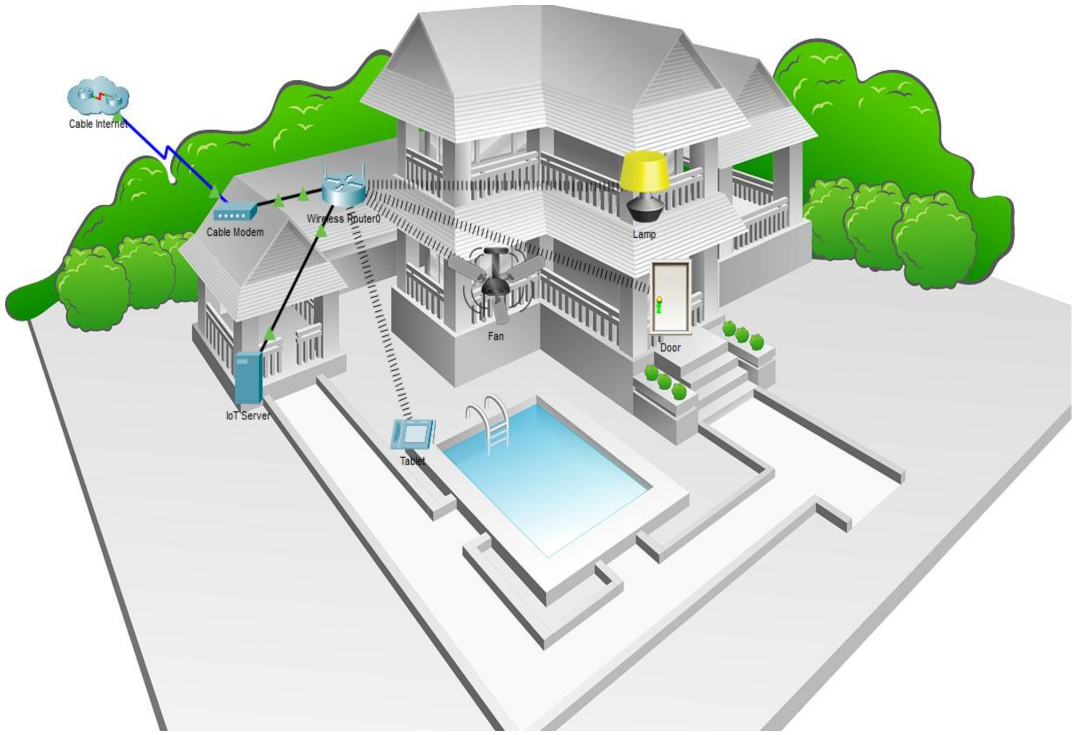


MODULE 2: ENVIRONMENTAL CONTROLS AND CUSTOM IOT DEVICES

Module 2 expanded upon smart home networking by introducing the Packet Tracer Physical Workspace — a simulated environment in which containers maintain independent environmental settings. Devices within these containers can both influence and respond to environmental variables such as ambient temperature, water level, and smoke concentration, enabling more realistic IoT behavior modeling.

2.0 Environmental Controls in Packet Tracer

The opening activity of this module focused on exploring and modifying environmental control parameters within Cisco Packet Tracer's Physical Workspace. Using the Environment window, ambient temperature settings were examined in relation to connected device behavior. Observations were made regarding how temperature fluctuations over time triggered corresponding responses in specific IoT devices, demonstrating the event-driven nature of smart home systems.

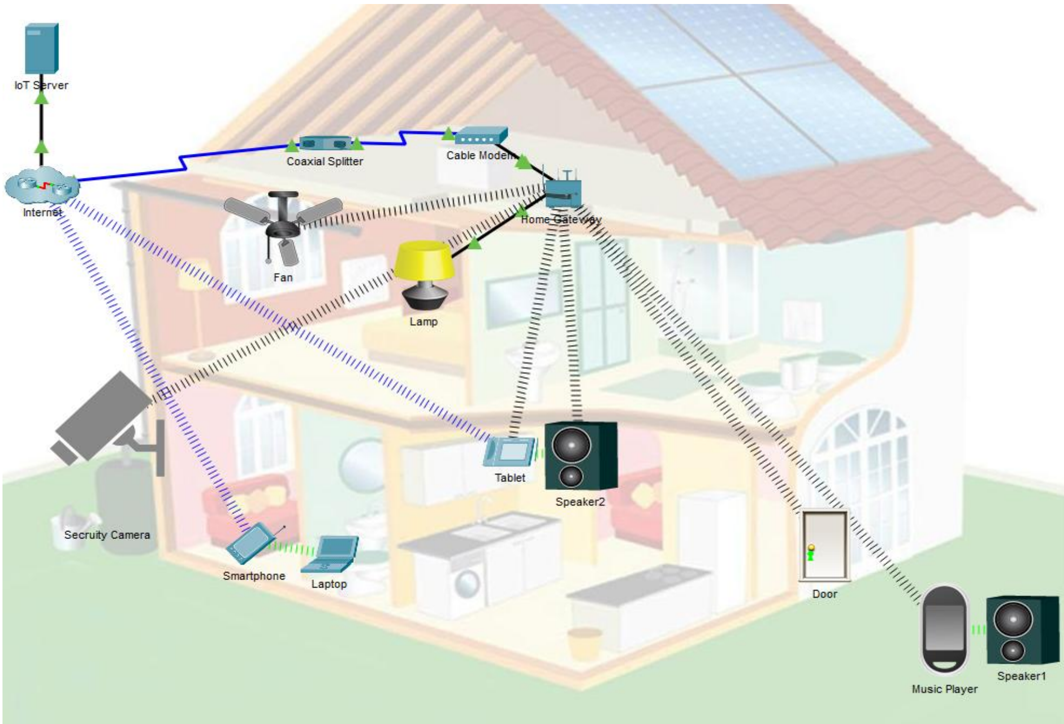


2.1 Create an IoT Thing in Packet Tracer

This section provided instruction on creating custom IoT devices — referred to as "things" — within Packet Tracer. The process included defining the device's visual states through graphic assets, reviewing pre-existing device scripts, and modifying those scripts to implement custom behavior. A supporting video walkthrough demonstrated the full device creation workflow before the hands-on activity commenced.

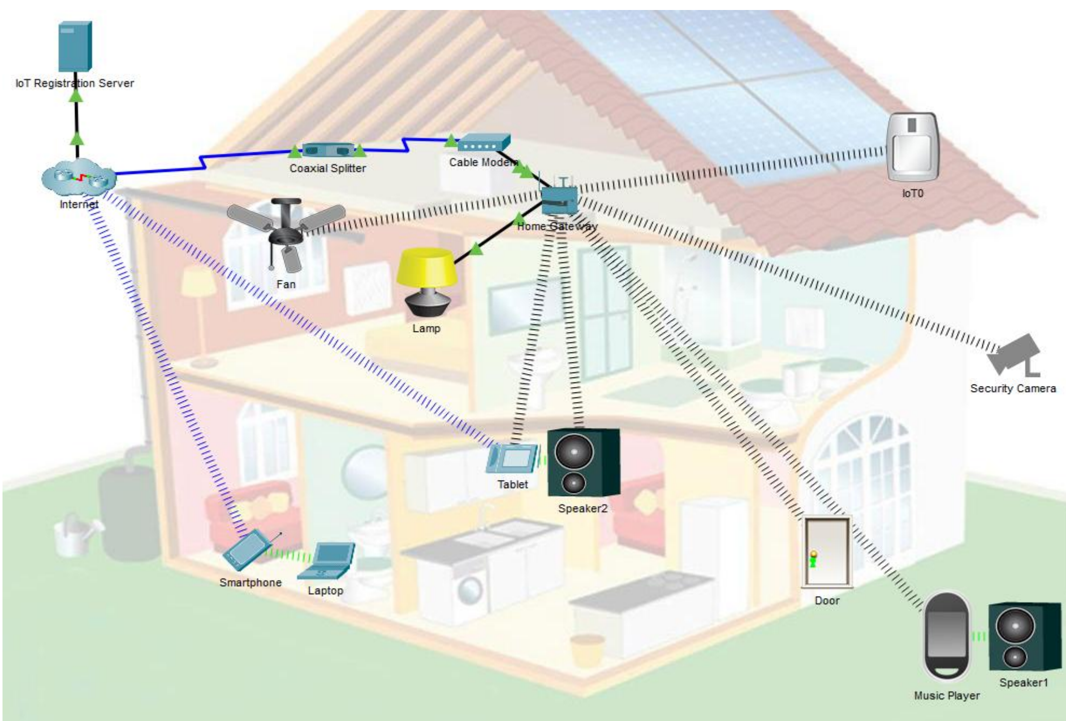
2.1.3 Create Your Own IoT Thing

In this activity, a custom IoT device was created from scratch within Cisco Packet Tracer. The device selected for implementation was a security camera. Because the provided image asset was unavailable for download, an alternative image of equivalent resolution and visual style was sourced independently. Following asset substitution, the device was renamed, assigned a wireless network adapter, and connected to the home gateway. Adapter configuration was completed and the wireless connection was verified successfully.



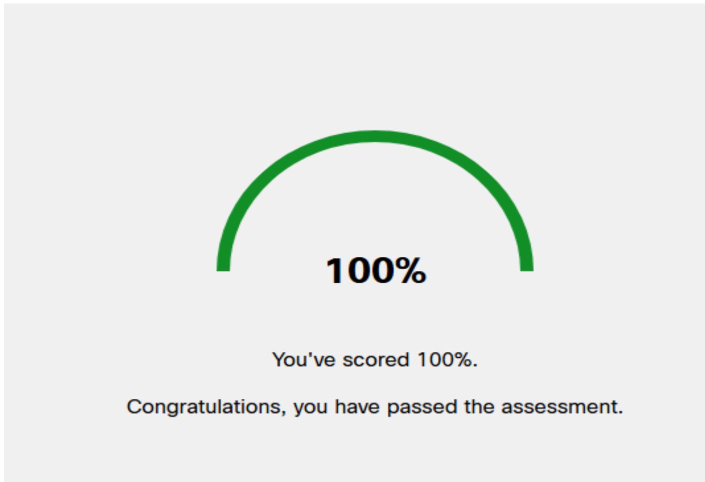
2.1.6 Modify an Existing Script for an IoT Thing

The final activity of the course involved modifying the script of an existing IoT device to introduce new functionality. The previously created security camera device was selected for modification. A secondary image asset — depicting a red flashing indicator light — was added to represent the camera's motion-activated state. The behavioral script was adapted from an existing motion sensor device; the relevant JavaScript code was copied, pasted into the security camera's script editor, and a single conditional line was updated to reflect the camera's response logic. The device was then connected to the IoT registration server, and motion-detection functionality was tested and confirmed to be operating as intended.



COURSE FINAL ASSESSMENT

Upon completion of all module activities, the course final assessment was administered. The assessment evaluated conceptual understanding and practical knowledge of IoT device configuration, wireless networking, environmental controls, and script modification within Cisco Packet Tracer.



CONCLUSION

This lab provided a comprehensive introduction to IoT device deployment and management within simulated smart home environments using Cisco Packet Tracer. Throughout the activities, core skills were developed in wireless device configuration, registration server management, environmental monitoring, custom device creation, and script-level programming.

The primary challenge encountered during the lab involved sourcing an appropriate image asset for the custom security camera device, as the original file provided in the activity could not be downloaded. This was resolved by independently locating a visually equivalent image of matching resolution. The workaround had no impact on the functional outcomes of the activity.

The competencies developed through this lab have practical applications in the growing field of smart home and IoT technology. As connected devices become increasingly prevalent in residential and commercial environments, the ability to configure, monitor, and extend IoT systems represents a valuable and transferable skill set.

REFERENCES

N/A

Generative AI Searches:

N/A

Collaboration:

This lab was completed independently.